Response to the reviewers

Common and Domain-Specific Design Patterns for Mobile AR Applications to Visualize Measures on Building Facades in an Environmental Context

Frank Fuchs-Kittowski, Maximilian Deharde, David Starzl, Marius Poppel, Simon Burkard

Thank you to both reviewers. The paper was accepted as a short paper. It is now 10 pages instead of 17 and uses the GI LNI template instead of LNCS. Below is what we did with each comment. Section numbers refer to the new version.

Reviewer 1 (overall 0, language 5)

Comment: The research question is presented as generally relevant, but the empirical basis is only two apps built by the authors themselves.

- We did not weaken this limitation. It is still in section 7.2 in full.
- Section 8 names further mAR applications for other facade measures as the next step.
- The two apps were picked because the domains are as far apart as possible while team, stack and process stayed the same. That is what makes it possible to separate shared from domain-specific patterns at all. It does not solve the problem of n = 2.

Comment: The limitations the authors list are truly significant.

- We agree, all four are still there in full (section 7.2).
- When the paper was cut by roughly 40 percent, this was one of the few passages that was not shortened at all.
- The missing user studies are the first item of further work in section 8.

Comment: "Design pattern" is used as a core concept without any definition or background, and the Context field should be phrased as a problem or challenge.

- The reviewer is right, and we find the point about the context field convincing. Phrasing it as a problem would fit both the pattern literature and the requirement wording we already use in sections 4 and 5.
- We could not fit it into this version. The paper sits at the 10 page limit of the short paper track. A grounded definition plus rewriting the seven Context fields needs about half a page and one or two additional references.

Comment: Nothing is said about the method used to identify the design patterns, and the inductive derivation of the eight dimensions has no methodological reference.

- Section 3 states that the work follows a comparative design study (SMM12), that the dimensions were derived inductively from the requirements documents of both prototypes, that they cover the development cycle from requirements to UX, and that they were agreed in a workshop with all authors.
- How we got from the dimensions to the patterns is not made explicit. That is a fair point.
- The page reduction made this worse rather than better: the table that mapped each pattern to the dimensions it came from had to be removed for space.
- Making the derivation explicit, with a reference to an established framework, is the first item on our list for an extended version.

Comment: A useful reflection on one development experience, but questionable as completed research.

- We would describe it the same way. It is a reflection based on two development projects, not a finished empirical study.
- The limitations in 7.2 and the further work in section 8 scope it accordingly.
- We think the short paper track is the right place for a contribution of this kind.

Reviewer 2 (overall 2, language 4)

Comment: Table 4 (D3) is internally inconsistent. The text says the PV app marks the whole area with a crosshair, but the Paradigm cell says a tap creates a discrete anchor point.

- Fixed. The seven per-dimension tables are now a single comparison table (Tab. 1, section 6.1).
- The D3 row now reads "Location-specific: one tap creates one anchor per plant object" for Facade-Greening-AR and "Area-continuous: corner points define an area that is filled with modules" for Facade-PV-AR.
- The row "Object Characteristics: Discrete / Discrete" was where the contradiction came from. It has been deleted, and the text in 6.1 now uses the same distinction.
- M5 and M6 were brought in line with this. M6 is now called "Continuous area coverage" everywhere, which is also what the abstract says.

Comment: The names of the two applications are used inconsistently across text and tables.

- Fixed throughout. It is Facade-Greening-AR and Facade-PV-AR in the running text, the comparison table, all figure and table captions and the section headings.

Other changes in the camera ready version

- Template changed from Springer LNCS to GI LNI, including the alphanumeric citation style ([Az97] instead of (Azuma 1997)) and the LNI caption conventions. DOIs were dropped because the LNI reference layout has no field for them.
- Shortened from 17 to 10 pages.
- The eight dimensions are presented in one table instead of seven. The text now discusses only D3, D4 and D5, where the substantive differences are.
- The discussion subsections Interpretation, Relationship between technical foundation and conceptual design, and Recommendations for practice were merged into section 7.1. The paragraph describing the structure of the paper and the pattern overview table were removed.
- Kept in full were all seven patterns M1 to M7, the method in section 3, all four limitations, both figures and all 20 references.
- Some smaller formal corrections